

Calculus on Manifolds Solutions

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June 2024

1 Measure Zero and Content Zero

A subset A of $\mathbb{R}^n$ has **measure 0** if for every $\varepsilon > 0$, there exists a cover $\{U_1, U_2, U_3, \dots\}$ of A by closed (or open) rectangles such that $\sum_{n=1}^{\infty} v(U_n) < \varepsilon$

A subset A of $\mathbb{R}^n$ has **content 0** if for every $\varepsilon > 0$, there exists a finite cover $\{U_1, U_2, U_3, \dots, U_n\}$ of A by closed (or open) rectangles such that $\sum_{i=1}^n v(U_i) < \varepsilon$

Theorem 3.5 If $a < b$, then $[a, b]$ does not have content 0. In fact, if $\{U_1, \dots, U_n\}$ is a finite cover of $[a, b]$, then $\sum_{i=1}^n v(U_i) \geq b - a$.

Proof. Say $U_1, \dots, U_n$ cover $[a, b]$ and let $a = t_0 < t_1 < t_2 < \dots < t_k = b$ be all endpoints of U_i . Then each $v(U_i)$ is the sum of some $(t_j - t_{j-1})$. Also, each $[t_{j-1}, t_j]$ is contained in some U_i . Thus, $\sum_{i=1}^n v(U_i) \geq \sum_{i=1}^k (t_i - t_{i-1}) = b - a \quad \square$

Exercises.

2 Integrable Functions

Lemma Let A be a closed rectangle and let $f : A \mapsto \mathbb{R}$ be a bounded function such that $\omega(f, x) < \varepsilon$ for all $x \in A$. Then, there is a partition P of A with $U(f, P) - L(f, P) < \varepsilon$.

Theorem 3.8 Let A be a closed rectangle and $f : A \mapsto \mathbb{R}$ a bounded function. Let $B = \{x \mid f \text{ is not continuous at } x\}$. Then f is integrable if and only if B has measure 0.

Partitions of Unity Exercises

3.37a Let

$$Q_n = [\frac{1}{2^{n+1}}, \frac{1}{2^n}] \quad \& \quad P_n = [1 - \frac{1}{2^n}, 1 - \frac{1}{2^{n+1}}]$$

Then,

$$(0, 1) = \bigcup_{i=1}^{\infty} (Q_i \cup P_i)$$

Let $\Phi \cup \Psi$ be a subordinate partition of unity to this open cover such that support $\phi_i = Q_i$ and support $\psi_i = P_i$. Then, $\int_{(0,1)} f$ exists if and only if

$$\sum_{i=1}^{\infty} \int_A (\phi_i f + \psi_i f)$$

converges. Therefore,

$$\begin{aligned} \int_A f &= \sum_{i=1}^{\infty} \int_A (\phi_i f + \psi_i f) \\ &= \lim_{N \rightarrow \infty} \int_A (\phi_i f + \psi_i f) \\ &= \lim_{N \rightarrow \infty} \int_{1/2^{N+1}}^{1-1/2^{N+1}} f \\ &= \lim_{\varepsilon \rightarrow 0} \int_{\varepsilon}^{1-\varepsilon} f \end{aligned}$$

We wrote $\varepsilon = 1/2^{N+1}$ in the last equality.

3.37b Let $A_n = [1 - \frac{1}{2^n}, 1 - \frac{1}{2^{n+1}}]$. Suppose that $f : (0, 1) \mapsto \mathbb{R}$ satisfies $\int_{A_n} f = (-1)^n/n$ and $f(x) = 0$ for $x \notin \text{any } A_n$. Show that $\int_{(0,1)} f$ does not exist, but $\lim_{\varepsilon \rightarrow 0} \int_{(\varepsilon, 1-\varepsilon)} f = \log 2$.

Proof. Let $\{\phi_1, \dots\}$ be a partition of unity having compact supports on $(0, 1)$ such that support $\phi_{2n} = A_n$ and support $\phi_{2n+1} \subset (0, 1/2]$ for all n . (We don't really care how ϕ_i behaves on $(0, 1/2)$ since $f = 0$ on that interval.) The integral $\int_{(0,1)} f$ exists if and only if the series

$$\sum_{i=1}^{\infty} \int_{(0,1)} \phi_i |f|$$

converges.

$$\begin{aligned} \sum_{i=1}^{\infty} \int_{(0,1)} \phi_i |f| &= \int_{(0,1)} \sum_{i=1}^{\infty} \phi_i(x) |f(x)| \\ &= \int_{(0,1)} |f(x)| \\ &= \int_{(0,1/2)} |f(x)| + \sum_{n=1}^{\infty} \int_{A_n} |f(x)| \\ &> \sum_{n=1}^{\infty} \left| \int_{A_n} f(x) \right| \\ &= \sum_{n=1}^{\infty} 1/n \rightarrow \infty \end{aligned}$$

Therefore $\int_{(0,1)} f$ does not exist. On the other hand, let N be such that $1/2^N > \varepsilon > 1/2^{N+1}$. The second part of the question is wrong. $\square$

Fubini's Theorem

3.23 Let $C \subset A \times B$ be a set of content 0. Let $A' \subset A$ be the set of all $x \in A$ such that $\{y \in B : (x, y) \in C\}$ is not of content 0. Show that A' is a set of measure 0.

Proof. Since C is of content 0, $\int_{A \times B} \mathbb{X}_C = 0$ by exercise 3-15. For each $x \in A$, let D_x denote the set $\{y \in B : (x, y) \in C\}$. Let $\varepsilon > 0$. If $x \notin A'$, there exist closed rectangles $U_1, \dots, U_k$ whose interior cover D_x and $\sum v(U_i) < \varepsilon$. Let P be a partition of B such that each subrectangle S of P is in one of the two groups:

$$\begin{aligned} S_1 &= \{S : S \cap D_x = \emptyset\} \\ S_2 &= \{S : S \subset U_i \text{ for some } 1 \leq i \leq k\} \end{aligned}$$

Then,

$$\begin{aligned} \sum_S M_S(\mathbb{X}_C) \cdot v(S) &= \sum_{S \in S_1} M_S(\mathbb{X}_C) \cdot v(S) + \sum_{S \in S_2} M_S(\mathbb{X}_C) \cdot v(S) \\ &= \sum_{S \in S_2} v(S) \\ &< \sum v(U_i) < \varepsilon \end{aligned}$$

Therefore, $U \int_B \mathbb{X}_C(x, y) dy = 0$. Meanwhile, for $x \in A'$, D_x is not of content 0, therefore $\int_B \mathbb{X}_C(x, y) dy \neq 0$. By Fubini's Theorem,

$$0 = \int_{A \times B} \mathbb{X}_C = \int_A \left(U \int_B \mathbb{X}_C(x, y) dy \right) dx$$

Then, A' is the set of discontinuities of the function

$$g(x) = U \int_B \mathbb{X}_C(x, y) dy$$

and has measure 0. $\square$

3.25 Prove by induction that $[a_1, b_1] \times \dots \times [a_n, b_n]$ does not have content 0 if $a_i < b_i$.

Proof. Let $U = [a_1, b_1] \times \dots \times [a_{n-1}, b_{n-1}]$. By Fubini's Theorem,

$$\int_R 1 = \int_{[a_n, b_n]} \left(\int_U 1 \right) dx = (b_n - a_n) \cdot \int_U 1 \neq 0$$

$\square$

3.26 Let $f : [a, b] \mapsto \mathbb{R}$ be integrable, non-negative. Let $A_f = \{(x, y) : a \leq x \leq b, 0 \leq y \leq f(x)\}$. Show that A_f is Jordan-measurable and has area $\int_a^b f$.

Proof. We have

$$L \int_R \mathbb{X}_{A_f}(x, y) dy = \int_0^{f(x)} 1 = f(x)$$

And by Fubini's Theorem,

$$\begin{aligned} \int_{A_f} 1 &= \int_a^b (L \int_R \mathbb{X}_{A_f}(x, y) dy) dx \\ &= \int_a^b f \end{aligned}$$

□

3.28 Give an easy proof of $D_{1,2}f = D_{2,1}f$ if both are continuous.

Proof. If $D_{1,2}f(t) - D_{2,1}f(t) \neq 0$ for some t , we can assume it is positive at t . By continuity, it is also positive around some closed rectangle $A = [a, b] \times [c, d]$ containing t . However, by Fubini's Theorem,

$$\begin{aligned} \int_A D_{1,2}f - D_{2,1}f &= \int_A D_{1,2}f - \int_A D_{2,1}f \\ &= \int_a^b \int_c^d D_{1,2}f(x, y) dy dx - \int_a^b \int_c^d D_{2,1}f(x, y) dy dx \\ &= \int_a^b (D_1f(x, d) - D_1f(x, c)) dx - \int_c^d \int_a^b D_{2,1}f(x, y) dy dx \\ &= f(b, d) - f(b, c) - f(a, d) + f(a, c) - \int_c^d (D_2f(b, y) - D_2f(a, y)) dy \\ &= f(b, d) - f(b, c) - f(a, d) + f(a, c) - (f(b, d) - f(a, d) - f(b, c) + f(a, c)) \\ &= 0 \end{aligned}$$

Notice that by continuity we were able to change the order of the iterated integrals. □

3.29 Use Fubini's Theorem to derive an expression for the volume of a set of $\mathbb{R}^3$ obtained by revolving a Jordan-measurable set in the yz -plane around the z axis.

Proof. Let C be the Jordan-Measurable set. Then $C \subset A$, where $A = [a, b] \times [c, d]$ is a closed rectangle in the yz -plane. Let D denote the set of $\mathbb{R}^3$ obtained by revolving C around the z -axis. Then, by Fubini's Theorem,

$$\int_D 1 = \int_c^d \int_a^b (2\pi y) \mathbb{X}_C(y, z) dz dy$$

□

3.32 Let $f : [a, b] \times [c, d] \mapsto \mathbb{R}$ be continuous and suppose $D_2 f$ is continuous. Define $F(y) = \int_a^b f(x, y) dx$. Prove Liebniez Rule: $F'(y) = \int_a^b D_2 f(x, y) dx$

Proof.

$$F(y) = \int_a^b \left(\int_c^y D_2 f(x, t) dt + f(x, c) \right) dx \quad (1)$$

$$\Rightarrow F'(y) = \left(\int_a^b \int_c^y D_2 f(x, t) dt dx \right)' \quad (2)$$

$$= \left(\int_c^y \int_a^b D_2 f(x, t) dx dt \right)' \quad (3)$$

$$= \int_a^b D_2 f(x, y) dx \quad (4)$$

□

Inverse Function Theorem.

2.37 Assume f is 1-1. Then, Df_1, Df_2 can't be constant or equal to 0. Then, there exists $(x, y) \in \mathbb{R}^2$ such that $Df_1(x, y) \neq 0$. Since f is continuously differentiable, there exists a neighbourhood A of (x, y) such that $Df_1(a, b) \neq 0$ for all $(a, b) \in A$. Consider the function

$$g : \mathbb{R}^2 \mapsto \mathbb{R}^2, g(x, y) = (f(x, y), y)$$

Notice,

$$Dg(x, y) = \begin{bmatrix} Df_1(x, y) & Df_2(x, y) \\ 0 & 1 \end{bmatrix}$$

and $\det Dg = Df_1(x, y) \neq 0$. By the IFT, there exists a neighbourhood V of (x, y) and W of $g(x, y)$ such that $g : V \mapsto W$ has a continous inverse. Now, write $f(x, y) = c$. Then, since W is open, there is a open set B containing $g(x, y) = (f(x, y), y)$ and contained in W . Take a point $(c, y') \in B$ where $y' \neq y$. Then, there exists $(x', y') \in g^{-1}(B)$ such that

$$g(x', y') = (c, y') \implies f(x', y') = c$$

but

$$f(x', y') = c = f(x, y)$$

a contradiction.

Better solution: Consider $f(0, y)$ and $f(x, 0)$ on $[-1, 1]$. Both are continuous on this interval, since $[-1, 1]$ is connected, the images of it under $f(0, y)$ and $f(x, 0)$ are connective as well. $f(0, 0)$ being a minimum or maximum of either function's image contradicts with injectivity of f . But then $g_x([-1, 1])$ and $g_y([-1, 1])$ both contain an open set around $f(0, 0)$, again, contradicting injectivity of f . ($g_x(a) = f(a, 0), g_y(a) = f(0, a)$)

Implicit Function Theorem 2.41 Define

$$h : R^2 \rightarrow R$$

$$h(x, y) = g'_x(y) = D_2 f(x, y)$$

Then,

$$Dh(x, y) = [D_{2,1}f(x, y) \quad D_{2,2}f(x, y)]$$

We are given $h(c, c(x)) = 0, \forall x$ and $c(x)$ is unique. Since $D_{2,2}f(x, y) \neq 0$ for all (x, y) , by the implicit function theorem, there exists an open sets A around x and B around $c(x)$ such that

$$\forall a \in A, \exists! k(a) \in B; h(a, k(a)) = 0$$

Since $c(x)$ is unique, we see that $k(a) = c(a)$. Then, c is differentiable, by IFT. Define

$$t : R \rightarrow R^2$$

$$x \mapsto (x, c(x))$$

Since $h(c, c(x)) = 0$, its derivative is the 0 function. Therefore,

$$D_{2,1}f(x, y) = D_{2,2}f(x, y) = 0$$

Notice that

$$\begin{aligned} 0 &= D_1 h(x, c(x)) = D(h \circ t)(x) \\ &= Dh(t(x)) \cdot Dt(x) \\ &= [D_{2,1}f(x, c(x)) \quad D_{2,2}f(x, c(x))] \cdot \begin{bmatrix} 1 \\ c'(x) \end{bmatrix} \end{aligned}$$

We see that

$$c'(x) = -\frac{D_{2,1}f(x, c(x))}{D_{2,2}f(x, c(x))}$$

Basic Definition of Integration

3.1 Take the partition $P_1 = \{0, 1/2, 1\}$ of $[0, 1]$ and let P_2 be any other partition of $[0, 1]$ Let $P = (P_1, P_2)$ be a partition of $[0, 1] \times [0, 1]$ Then,

$$L(f, P) = 1/2 = U(f, P)$$

3.2 Let $\epsilon > 0$. Assume $f(x) \neq g(x)$ at points $x = x_1, \dots, x_n$ Let P be a partition of A where the subrectangles containing x_i have area less than $\frac{\epsilon}{n \max g(x_i)}$

3.3 Let P be a partition of A , S a subrectangle of P . For all $a \in A$,

$$\inf f(x) + \inf g(x) \leq f(a) + g(a)$$

Taking the infimum of both sides,

$$m_S(f) + m_S(g) = \inf f(x) + \inf g(x) \leq \inf(f+g)(x) = m_S(f+g)$$

The rest follows.

3.4 Assume f is integrable. Let P be a partition of A . Let $\epsilon > 0$ be given. There exists a partition P' such that

$$U(f, P') - L(f, P') < \epsilon$$

Let P_1 be the refinement of P and P' . Then, the above inequality holds for P_1 as well. Let S be a subrectangle of P . Then, there exists a set $\mathbb{S}$ of subrectangles of P_1 such that

$$S = \bigcup \mathbb{S}$$

That is, $\mathbb{S}$ is a partition of S into smaller rectangles. Now,

$$\epsilon > U(f, P_1) - L(f, P_1) = \sum U(f, \mathbb{S}') - L(f, \mathbb{S}') > U(f, \mathbb{S}) - L(f, \mathbb{S})$$

where $\mathbb{S}'$ is the partition of the rectangle S' in P . So, S is integrable. Other direction follows immediately if you think about it for a second.

3.6 Assume f is integrable. Write $|f| = f^+ + f^-$ where $f^+ = \max\{0, f\}$, $f^- = \max\{0, -f\}$.

Measure 0 and Content 0

3.9 a). For any open cover O of A , if A is not bounded, it is not compact, and there does not exist a finite subcover. Therefore A cannot have content 0. b). Take the set of all integers.

3.10 a.) Let $U_1, \dots, U_n$ cover C such that $\sum v(U_i) < \epsilon$. Since each U_i is closed, $\overline{C} \subset \bigcup U_i$. Then, the boundary is also covered. b.) Take Q . Boundary Q is $[0, 1]$ and does not have measure 0.

3.11 measure $bdA = \text{measure } [0, 1] - \text{measure } A \neq 0$. Or prove by contradiction. Let $\sum(b_i - a_i) = t < 1$. If bdA has measure 0 then $\exists U_i$ such that $\bigcup_i U_i \supset bdA$

$$\sum v(U_i) < 1 - t$$

Then, $[0, 1] \setminus U_i$ is contained in A . So

$$t = \sum_i (b_i - a_i) \geq \sum_i v([0, 1] \setminus U_i) > t$$

contradiction.

3.12 From problem 1.30, we know that for increasing $f : [1, b] \rightarrow R$,

$$\sum_{i=1}^n o(f, x_i) < f(b) - f(a) \quad (1)$$

for distinct points $x_1, \dots, x_n \in [a, b]$. Now,

$$\{x | f \text{ is discontinuous at } x\} = \bigcup_n B_n$$

where $B_n = \{x | o(f, x) > \frac{1}{n}\}$. Each B_n is finite because otherwise we can find adequate amount of x_i such that equation (1) is violated. Then, $\bigcup B_n$ has measure 0.

3.13 a.) By induction on n . Assume $P(n-1)$. Then, let $[a, b] = [a_1, b_1] \times \dots \times [a_{n-1}, b_{n-1}]$. The rectangles $[a, b]$ can be arranged in a sequence $(t_1, t_2, \dots)$. Similarly, the rectangles $[a_n, b_n]$ can be arranged in a sequence $(s_1, s_2, \dots)$ because the statement is true for $n = 1$. Now, we can arrange $t_i \times s_j$ in a sequence as well, because the theorem is true for $n = 2$.

b.) Let O be an open cover, using the hint, for each $x \in A$, we find a rectangle B_x such that $x \in B_x \subset U_x$ for some $U_x \in O$. Then, take these U_x . There is surjective function from $\{B_x | x \in A\}$ to $\{U_x | x \in A\}$ sending B_x to U_x . Since the domain is at most countable by part a, we can get the desired sequence of U_i .

Integrable Functions

Lemma 3.7 For each $x \in A, \exists U_x \subset A; x \in U_x$ & $M_{U_x}(f) - m_{U_x} < \epsilon$. Since A is compact, finitely many $\{U_x\}$ cover A . Let P be a partition such that each subrectangle S is contained in some U_x .

Theorem 3.8

Proof. $\Leftarrow$ direction.

Let $\epsilon > 0$ and $|f| < M$. Let $B = \{x \in A | o(f, x) > \epsilon\}$. Since B is a subset of the set of all discontinuities. Therefore, has measure 0. Since A is compact, B is content 0. Let

$$U_1 \cup \dots \cup U_n \supset B$$

and

$$\sum v(U_i) < \epsilon/2M$$

Let P be a partition of A such that each subrectangle S is either contained in some U_i or has empty intersection with all U_i . Define

$$\begin{aligned} \mathcal{S}_1 &= \{S | S \cap B = \emptyset\} \\ \mathcal{S}_2 &= \{S | S \cap B \neq \emptyset\} \end{aligned}$$

Then, by lemma for each $S \in \mathcal{S}_1$, there exists a partition of S , P_S such that

$$U(f, P_S) - L(U, P_S) < \epsilon \cdot v(S)$$

Let P' be the partition that refines P in a way that each subpartition P_S for $S \in \mathcal{S}_1$ is contained in P' . Then,

$$\begin{aligned} U(f, P') - L(f, P') &= \sum_{S' \subset S \in \mathcal{S}_1} [M_{S'}(f) - m_{S'}(f)]v(S') + \sum_{S \in \mathcal{S}_2} [M_S(f) - m_S(f)]v(S) \\ &< \epsilon \cdot \sum_{S \in \mathcal{S}_1} v(S) + 2M \sum_{S \in \mathcal{S}_2} v(S) \\ &< \epsilon v(A) + 2M \cdot \frac{\epsilon}{2M} \\ &= \epsilon \cdot (v(A) + 1) \end{aligned}$$

So, f is integrable. □

3.16 Take $C = \{q \in \mathcal{Q} | 0 \leq q \leq 1\}$.

3.17 b.d. C and $\overline{C}$ have measure 0. Therefore, $\overline{C}$ also has measure 0. Since C is bounded, $\overline{C}$ is bounded and closed. Therefore, $\overline{C}$ is compact, and, thus, of content 0. By problem 3.15 $\int_A \mathcal{X}_{\overline{C}} = 0$ and

$$\int_A \mathcal{X}_C \leq \int_A \mathcal{X}_{\overline{C}} = 0$$

3.18 Let $B_n = \{x | f(x) > 1/n\}$. Then,

$$B = \{x | f(x) \neq 0\} = \bigcup_n B_n$$

There exists a partition P of A such that

$$U(f, P) < \epsilon/n$$

Then, let $\mathcal{S}_1 = \{S | S \cap B_n = \emptyset\}$, $\mathcal{S}_2 = \{S | S \cap B_n \neq \emptyset\}$. Then,

$$\begin{aligned} U(f, P) &= \sum_S M_S(f) \\ &= \sum_{S \in \mathcal{S}_1} M_S(f) + \sum_{S \in \mathcal{S}_2} M_S(f) \\ &< 1/n \cdot \sum_{S \in \mathcal{S}_2} v(S) \\ &\implies \epsilon > \sum_{S \in \mathcal{S}_2} v(S) \supset B_n \end{aligned}$$

Thus, each B_n has content 0 and, therefore, B has measure 0.

3.21 Let P be such a partition. Notice that $S_2 \subset S_1$ and $S_1 \setminus S_2$ covers boundary C . Also notice that,

$$\sum_{S \in S_1 \setminus S_2} v(S) = \sum_{S \in S_1} v(S) - \sum_{S \in S_2} v(S) < \epsilon$$

This shows C is Jordan-measurable.

Algebraic Preliminaries

4.3 Let $v_1, \dots, v_n$ be an ON basis wrt direct product T such that $[v_1, \dots, v_n] = \mu$. Write

$$w_i = \sum_{j=1}^n a_{ij} v_j$$

By theorem 4-6

$$w(w_1, \dots, w_n) = \det(a_{ij})w(v_1, \dots, v_n)$$

Since w is the volume element, $w(w_1, \dots, w_n) = \det(a_{ij})$. Let $A = (a_{ij})$. The hint reveals that

$$g_{ij} = \sum_{k=1}^n a_{ik}a_{jk} = (AA^T)_{ij}$$

Then,

$$\begin{aligned}\det(g_{ij}) &= \det(AA^T) \\ &= \det A \cdot \det A^T \\ &= (\det A)^2 \\ &= |w(w_1, \dots, w_n)|^2\end{aligned}$$

as desired.

4.5 Let e_i be the normal basis. Write

$$c^i(t) = \sum a_{ij}^t e_j$$

Since each $(c^1(t), \dots, c^n(t))$ is a basis for $\mathbb{R}^n$, and $\det$ is a continuous function and c is continuous, if $\det(a_{ij}^0) > 0$, then there is a neighbourhood $[0, \delta]$ around 0 such that for $t \in [0, \delta]$,

$$\det \circ c(t) = \det(a_{ij}^t) > 0$$

and so

$$[c^1(0), \dots, c^n(0)] = [c^1(t), \dots, c^n(t)]$$

Since $[0, 1]$ is compact, there exists finitely many intervals such that this process can be carried and we can conclude

$$[c^1(0), \dots, c^n(0)] = [c^1(1), \dots, c^n(1)]$$

Fundamental Theorem of Calculus

4-25 We will use the change of variables theorem (Theorem 3-13) and Theorem 4-9. We know

$$c^*w(x) = f(x)dx^1 \wedge \dots \wedge dx^k$$

Then,

$$\begin{aligned}\int_c w &= \int_{[0,1]^k} c^*w \\ &= \int_{[0,1]^k} f\end{aligned}$$

On the other hand, we have

$$(c \circ p)^* w = p^*(c^* w) = p^*(f dx^1 \wedge \cdots \wedge dx^k) = (f \circ p) \det p' dx^1 \wedge \cdots \wedge dx^k$$

So,

$$\begin{aligned} \int_{c \circ p} w &= \int_{[0,1]^k} (c \circ p)^* w \\ &= \int_{[0,1]^k} p^* c^* w \\ &= \int_{[0,1]^k} (f \circ p) \det p' \\ &= \int_{p([0,1]^k)} f \quad (\text{using CoV since } \det p' = |\det p'| \text{ is given}) \\ &= \int_{[0,1]^k} f \\ &= \int_c w \end{aligned}$$

as desired. Note also that to be able to use change of variables, we need the open set $(0,1)^k$ rather than the closed set. However, since the boundary of $[0,1]^k$ has measure 0, the value of the integral does not change so we can proceed by applying the change of variables on $(0,1)^k$.

4-28 Computing $c_{(1,0)} c_{c(1,1)}$ both give $t(R,0) + (1-t)f(R,0)$ so they cancel each other out. Then

$$\partial c = c_{(2,0)} - c_{(2,1)} = c_{R,f}(t) - c_{R,n}(t)$$

For the second part, the errata fix tells us to take $c_{R^n,n}$ instead of $c_{R,n}$. We do so. First, we show $c(0,t) \neq 0$ for large enough R .

$$\begin{aligned} c(0,t) = t(R^n,0) + (1-t)f(R) = 0 &\iff f(R) = \frac{-t}{1-t}R \\ &\iff \left(\frac{1+t}{1-t}\right)R^n + a_1 R^{n-1} + \cdots + a_{n-1}R + a_0 = 0 \end{aligned}$$

For big enough R , the last expression tends to infinity and therefore $c(0,t) \neq 0$. For an arbitrary point (x,y) on the circle with radius R , the corresponding point on the circle with radius R^n is $(R^{n-1}x, R^{n-1}y)$. Notice that if $c_{R^n,n}(s,t) = (R^{n-1}x, R^{n-1}y)$ then

$$c(s,t) = t(R^{n-1}x, R^{n-1}y) + (1-t)f(x,y) = 0 \iff (x,y)^n + \cdots + a_{n-1}(x,y) + a_n = -\frac{t}{1-t}(R^{n-1}x, R^{n-1}y)$$

The last equation is preserved if we apply a change of coordinates operation and change to the suitable coordinates that sends (x,y) to $(R,0)$ and

$(R^{n-1}x, R^{n-1}y)$ to $(R^n, 0)$. Then we can argue in a similar way to the $c(0, t)$ case to show that for big enough R , the above equation does not hold for all $t \in [0, 1]$.

b) COMING SOON!

4-29 We integrate as suggested by the hint.

$$\begin{aligned}\int_{[0,1]} w - \lambda x &= \int_{[0,1]} f(x) dx - \lambda \int_{[0,1]} dx \\ &= \int_0^1 f - \lambda\end{aligned}$$

On the other hand,

$$\int_{[0,1]} dg = \int_{\partial[0,1]} g = g(1) - g(0) = 0$$

Since we want $w - \lambda dx = dg$, we choose

$$\lambda = \int_0^1 f(x) dx$$

4-32 Assume $w = d\eta$, ie w is exact. Then, by Stokes's theorem,

$$\begin{aligned}\int_{c_1} w &= \int_{\partial c_1} \eta \\ &= \eta(c_1(1)) - \eta(c_1(0)) \\ &= \eta(c_2(1)) - \eta(c_2(0)) \\ &= \int_{\partial c_2} \eta \\ &= \int_{c_2} w\end{aligned}$$

as desired. To prove existence of c , simply take the homotopy from c_1 to c_2 . That is define $c : I \times I \rightarrow \mathbb{R}^2$ by

$$c(s, t) = tc_2(s) + (1 - t)c_1(s)$$

Then $c_3 = c_1(1)$, $c_4 = -c_1(0)$ and $\partial c = c_1 - c_2 + c_3 - c_4$ as desired.

b) Let $c : I \times I \rightarrow \mathbb{R}^2$ be given by $c(t) = tc_1(s) + (1 - t)c_2(s)$. Then $\partial c = c_1 - c_2$ and by stokes theorem and by exercise 4-31

$$0 = \int_{\partial c} w = \int_c dw \implies dw = 0$$

Since w is a 1-form, $w = h \cdot dx + g \cdot dy$ and since w is closed, $D_2 h = D_1 g$. Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \int_0^x h(t, 0) dt + \int_0^y g(x, t) dt$$

By problem 2-21 and 3-34 we see that $D_1f = h, D_2f = g$ and therefore

$$w = D_1f dx + D_2f dy = df$$

showing that w is exact.

4-30 First of all, the 1-form $c_{R,1}^*w$ satisfies the requirements in exercise 4-29 and therefore such λ_R and g_R assumed in the hint do exist. Define $c : I \times I \rightarrow \mathbb{R}^2 - 0$ by $c(s, t) = tc_{R_1,1}(s) + (1-t)c_{R_2,1}(s)$. Then, $\partial c = c_{R_2,1} - c_{R_1,1}$. Since $dw = 0$,

$$\begin{aligned} 0 &= \int_c dw \\ &= \int_{\partial c} w \\ &= \int_{c_{R_1,1}} w - \int_{c_{R_2,1}} w \\ &= \lambda_{R_1} - \lambda_{R_2} \end{aligned}$$

The last equation is justified in the solution of 4-29. Then, we can write

$$w = \frac{\lambda}{2\pi} d\theta + dg$$

where $\lambda = \int_{c_{R,1}} w$ for an arbitrary $R > 0$ and $g(x) = g(0) + \int_{[0,x]} w - \frac{\lambda}{2\pi} d\theta$. Questions: What is g exactly? Can I assume $g(0) = 0$? How does the switch from dx to $d\theta$ work? Right now, it feels like dark magic.

4-34 Easy.

4-33 a) It is easy to check $f(z) = z$ is analytic. For $f(z) = \bar{z}$, we see that

$$\begin{aligned} \lim_{z \rightarrow z_0} \frac{\bar{z} - \bar{z}_0}{z - z_0} &= \lim_{z \rightarrow z_0} \frac{(a - a_0) - (b - b_0)i}{(a - a_0) + (b - b_0)i} \\ &= \lim_{\tilde{a}, \tilde{b} \rightarrow 0} \frac{\tilde{a}^2 - 2\tilde{a}\tilde{b} - \tilde{b}^2}{\tilde{a}^2 + \tilde{b}^2} \end{aligned}$$

The last limit does not exist as the value for $(\tilde{a}, 0)$ is 1 whereas the value for $(0, \tilde{b})$ is -1. The sum, product and quotient are analytic and this can be proved easily.

Introduction to Manifolds

5-1 Consider $x \in \partial M$. Let $f : W \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$ be a coordinate chart such that $f(a) = x$. Let $g : W' \subset \mathbb{R}^k \rightarrow \mathbb{R}^n$ be another coordinate chart around $y \in \partial M$ whose image contains x . We need to prove x is a boundary point according to chart g too. That is, we need to show $c^k = 0$ for $c \in W'$ with $g(c) = x$. Consider the function $f^{-1} \circ g : g^{-1}(f(W)) \rightarrow W$. This function is continuous with its inverse also continuous. Moreover, $f^{-1} \circ g(c) = a$. We know

x is a boundary point with respect to the chart f , that is $a^k = 0$. If $c^k \neq 0$, then c is an interior point of W' . $f^{-1} \circ g$ is an open set (because its inverse is continuous). Then, it follows that a must be an interior point of W . However this is a contradiction since $W \subset \mathbb{H}^k$ and $a^k = 0$, ie a is in the boundary of W . Thus, $c^k = 0$ and x is a boundary point wrt to chart g as well. Then, it follows that if $x \in \partial M$, by the condition (M') there is a diffeomorphism $h : U \rightarrow V$ such that $h(U \cap M) = V \cap (\mathbb{H}^k \times \{0\})$ and $h(y) = 0$ for all $y \in U \cap \partial M$. Then, $h(U \cap \partial M) = V \cap (\mathbb{R}^{k-1} \times \{0\})$ proving that ∂M is a $k-1$ dimensional manifold. To see that $M - \partial M$ is a k - dimensional manifold, observe that if a point x is not a boundary point wrt a chart f , then it also not a boundary point wrt any other chart g whose image contains x .

5-2 We proceed as suggested in the hint. Let $\alpha : (0, 1) \rightarrow M$ be the continuous map that sends 0 to the starting point of the figure 6 and 1 to the junction point. α is 1-1 and continuous and differentiable and α' has rank 1. But α^{-1} is not continuous as around the junction point, the preimage of the open set around $x \in (0, 1)$ is not open - any open set around $\alpha(x)$ contains many points whose preimage is very close to 1. So condition 3 is not satisfied. Also, figure 6 is not a manifold because one cannot find a diffeomorphism around the junction point that is a copy of the real line.

5-3 If $x \in \text{int } N$ then let $U \ni x$ be open in $\mathbb{R}^n$, contained in $\text{int } N$. An open subset of $\mathbb{R}^n$ is an n -dimensional manifold. Thus, x satisfies the condition M . If $x \in N \setminus \text{int } N = \text{bd } N$, then $x \in \text{bd } A$ also. Since the latter is an $n-1$ dimensional manifold, there exists open sets $U \ni x$, V in $\mathbb{R}^n$ and a diffeomorphism $h : U \rightarrow V$ such that $h(U \cap \text{bd } A) = V \cap (\mathbb{R}^{n-1} \times \{0\})$. Let $\mathbb{H}_+^n = \{y \in V : \pm y^n > 0\}$. If there is a point $p \in V \cap \mathbb{H}_+^n$ then our claim is that $V \cap \mathbb{H}_+^n \subset h(U \cap N)$. Assume, for contradiction's sake, that there is $q \in \mathbb{H}_+^n \cap V \setminus h(U \cap N)$. Then the line segment $\overline{pq}$ contains a point of $\mathbb{H}_+^n$ that is a boundary point of the image of h . Since h is a diffeomorphism, this means the segment $\overline{pq}$ will give rise, through h^{-1} to a boundary point of $U \cap N$, thus to a boundary point of A . But the boundary points of A in U are in 1-1 correspondence with $V \cap (\mathbb{R}^{n-1} \times \{0\})$. So $h(U \cap N)$ is either $\mathbb{H}_+^n$, $\mathbb{H}_-^n$ or their union. It cannot be the union as then a point a in the intersection of V with the hyperplane $\{x \in \mathbb{R}^n | x^n = 0\}$ will become an interior point whereas it mapped to a boundary point of A via h . If $h(U \cap N) = \mathbb{H}_-^n$, then replace h with the function that reverses the sign of the n -th component of $h(x)$.

5-4 We know $\exists h : A \rightarrow V$ diffeomorphism. $A \ni x, V \subset \mathbb{R}^n$. We also know by theorem 2-13 that this h is actually the $F : \mathbb{R}^k \times \mathbb{R}^{n-k} \rightarrow \mathbb{R}^n$ given by $F(x, y) = (x, f(x, y))$ had M been defined as $f^{-1}(0)$. Taking this hint, we argue that $g = \pi_2 \circ h$ is the desired function. First we check

$$\begin{aligned} g(M \cap A) &= \pi_2 \circ h(M \cap A) \\ &= \pi_2(V \cap (\mathbb{R}^k \times \{0\})) \\ &= 0 \end{aligned}$$

Also $\det h' \neq 0$ because h is a diffeomorphism. Thus, $Dh(x)$ has rank n . Then,

$$Dg(x) = D\pi_2(h(x)) \circ Dh(x) = \pi_2 \circ Dh(x)$$

has rank $n - k$.

5-5 Let $A \subset \mathbb{R}^n$ be a k dimensional subspace of $\mathbb{R}^n$. Let $\{v_1, \dots, v_n\}$ be an ordered basis for A . Extend it to a basis $\{v_1, \dots, v_n\}$ of $\mathbb{R}^n$. Let $x \in A$. Take a neighbourhood U of x and define $h : U \rightarrow U$ by $h(v_i) = e_i$. Then, for $y \in U \cap A$, there exists $a_1, \dots, a_k$ with $y = \sum_{i=1}^k a_i v_i$. Then, $h(y) = \sum_{i=1}^k a_i e_i \in U \cap (\mathbb{R}^k \times \{0\})$. Note that h is a diffeomorphism since it is an isomorphism from $\mathbb{R}^n$ to itself.

5-6 Let $M = \{(x, y) | y = f(x)\}$. Let $(a, b) \in M$. Let $U \ni (a, b)$ be open in $\mathbb{R}^{n+m}$. Let $W = \{x \in \mathbb{R}^n | (x, f(x)) \in U \cap M\}$. If we define $g : W \rightarrow \mathbb{R}^{n+m}$ then automatically $g(W) = U \cap M$. Also, $g'(x)$ has rank n on W since the derivative matrix looks like $\begin{bmatrix} I_n \\ Df(x) \end{bmatrix}$. Furthermore $g^{-1} : g(W) \rightarrow W$ is basically the projection function onto the first n coordinates and thus continuous. Hence g is a coordinate chart for M , proving M is an n -dimensional manifold.

Fields and Forms on Manifolds

5-9 Let $c : I \rightarrow M$ be a curve with $c(0) = x$ and $f : W \rightarrow \mathbb{R}^n$ be a coordinate system around $x = f(a)$. Then, $f^{-1} \circ c(0) = a$. Consider the vector $Df^{-1}(x)(c'(0)) \in \mathbb{R}^k \cong \mathbb{R}_a^k$. The pushforward of this tangent vector is $Df(a) \circ Df^{-1}(f(a)) \circ c'(0) = D(f \circ f^{-1})(x)(c'(0)) = c'(0) \in M_x$ as desired. Conversely, assume $v \in M_x$. Then, $v = df_a(w)$ for some $w \in \mathbb{R}_a^k$. Write $w = (b_1, \dots, b_k) \in \mathbb{R}^k$. Let $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^k$ be given by $\gamma(t) = (a_1 + b_1 t, \dots, a_k + b_k t)$. (Choose ϵ so that the image of γ lies inside W .) Let $c = f \circ \gamma$. Then, $c(0) = f(a) = x$ and $c'(0) = Df(a)(\gamma'(0)) = Df(a)(w) = v$. Thus, M_x consists precisely of tangent vectors at 0 of the curves c in M with $c(0) = x$.

5-10 Let $x \in M$. Choose a coordinate system f around x from $\mathcal{C}$. Then let

$$\mu_x = [df_a(e_1), \dots, df_a(e_k)]$$

This defines an orientation μ on M . Let $g : W \rightarrow \mathbb{R}^n \in \mathcal{C}$. We will prove g is orientation preserving. Pick $x \in g(W)$. Say $g(b) = x$. There exists a coordinate system $f \in \mathcal{C}$ around $x = f(a)$. Then,

$$\begin{aligned} \mu_x &= [df_a((e_1)_a), \dots, df_a((e_k)_a)] = [dg_b((e_1)_b), \dots, dg_b((e_k)_b)] \\ &\iff [d(g^{-1} \circ f)_a(e_1), \dots, d(g^{-1} \circ f)_a(e_k)] = [(e_1)_b, \dots, (e_k)_b] \end{aligned}$$

Since $\det d(g^{-1} \circ f) = \det(g^{-1} \circ f)' > 0$ is given, we see that g is orientation preserving.

5-11 Let $n(x) = df_0(v)$ where $v^n < 0$ be the outward unit normal at $x \in \partial M$. Then, the induced orientation on ∂M is $[(v_1, \dots, v_{n-1})]$ where

$$[(n(x), v_1, \dots, v_{n-1})] = [(e_1, \dots, e_n)]$$

Now, think of ∂M as the main $(n - 1)$ -dimensional manifold and apply the latter definition. Let $n'(x)$ be the unit vector perpendicular to $(\partial M)_x$ satisfying

$$[(n'(x), v_1, \dots, v_{n-1})] = [(e_1, \dots, e_n)]$$

Since both $n(x), n'(x)$ are perpendicular to $(\partial M)_x$, and unit vectors, either $n(x) = n'(x)$ or $n(x) = -n'(x)$. It is obvious that the former must hold by

$$[(n'(x), v_1, \dots, v_{n-1})] = [(e_1, \dots, e_n)] = [(n(x), v_1, \dots, v_{n-1})]$$

5-12 Let F be a differential vector field on M . Then on any chart $(U_\alpha, \phi) = (U, x_\alpha^1, \dots, x_\alpha^n)$, $F = \sum a_\alpha^i \frac{\partial}{\partial x_\alpha^i}$ for smooth functions $a_\alpha^i : U_\alpha \rightarrow \mathbb{R}$. Let $b_\alpha^i : \tilde{U}_\alpha \rightarrow \mathbb{R}$ be a smooth extension. Then let $\{\rho_\alpha\}$ be a compactly supported partition of unity subordinate to the open cover $\{U_\alpha\}$ of M and let

$$\tilde{F} = \sum_n \sum_i \rho_\alpha \cdot b_\alpha^i \frac{\partial}{\partial x_\alpha^i}$$